

- 7 (a) (i) Faraday's law of electromagnetic induction states that the magnitude of the induced emf in a coil is directly proportional to the rate of change of magnetic flux linkage of the coil. [1]

Lenz's Law states that the direction of induced e.m.f. is such that its effects oppose the change which causes it. [1]

- (ii) When the card is brought in range of a card reader, the varying (or sinusoidal) magnetic flux density / field generated by the card reader results in a changing magnetic flux linkage through the loops/coils of printed wire embedded in the card. [1]

By laws of electromagnetic induction, there is an emf induced in the coils (which is in the direction to oppose the varying magnetic field). [1]

The closed circuit allows the emf to induce a current (and thus supplies power to the internal transmitter, which can then send its data to the card reader). [1]

(iii) 1. $\varepsilon = -\frac{d\phi}{dt} = -\frac{d(NBA \sin 2\pi ft)}{dt} = -2\pi f NBA \cos 2\pi ft$ [1]
 Peak e.m.f. $\varepsilon_o = 2\pi f NAB_o = 2\pi (13.56 \times 10^6) (3) (4.0 \times 10^{-3}) B_o$ [1]
 $= 1.02 \times 10^6 B_o$ [1]

2. $V_{rms} = V_o / \sqrt{2}$ [1]
 $10.0 \times 10^{-3} = 1.02 \times 10^6 B_o / \sqrt{2}$
 $B_o = 1.38 \times 10^{-8} \text{ T}$ [1]

- (iv) The alternating magnetic field from machine will still induce an emf with either face of the card placed on machine / Alternating current will flow regardless of which face is used to tap machine. [1]

- (b) The root-mean-square value of an alternating current is the value / magnitude of a steady direct current that will dissipate energy at the same (average) rate as the alternating current in a given resistance. [1]

- (c) (i) Alternating voltage can be stepped up or down easily. [1]

(ii) Step-up transformer with turns ratio $\frac{N_s}{N_p} = 240/120 = 2.0$ [1]

(iii) Assuming no power losses in the transformer, [1]
 $P = IV$
 $1000 = 120 I_p$
 $I_p = 8.33 \text{ A}$ [1]

(iv) 1. By conservation of energy,
 Power supplied to primary coil
 $=$ Power output from secondary coil + power loss in windings of transformer
 $= 1000 + 600 = 1600 \text{ W}$ [1]

Output voltage of transformer $V_s = 240 \text{ V}$
 Secondary current $= 1000 / 240$ [1]
 $= 4.17 \text{ A}$ [1]

2. efficiency $= 1000/1600 \times 100\% = 62.5 \%$ [1]